AHMED ESAM YAHIA HALLAWA

Mechatronics & AI Automation Engineering Student

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PROFESSIONAL SUMMARY

Fourth-year Mechatronics Engineering student at Benha University specializing in Systems and AI Automation. Builds end-to-end intelligent systems — from YOLOv8 detection and persistent re-identification pipelines to PID-controlled mobile robots and microcontroller firmware — with a focus on real-time, multithreaded, production-grade software. Brings 1.5+ years of remote experience training and evaluating large language models (60+ projects) and a proven leadership record as IEEE Student Branch Treasurer, managing event budgets and screening 750+ recruitment applicants.

EDUCATION

B.Sc. Mechatronics Engineering — Systems & AI Automation | Faculty of Engineering, Benha University — Expected 2027

An interdisciplinary program integrating mechanical design, electrical and electronic systems, control engineering, embedded programming and artificial intelligence to design, model and automate intelligent machines and robotic systems.

WORK EXPERIENCE

AI Trainer — LLM Evaluation & RLHF | Outlier AI (Remote) — Jan 2023 – Sep 2024

- Trained and improved large language models for client companies through structured human feedback, response ranking and review of model-generated scripts.
- Contributed to 60+ projects across markets including Egypt and Saudi Arabia, in multiple languages and under varied evaluation rubrics.

TECHNICAL PROJECTS

Persistent Re-ID Surveillance & Tracking System

Python · YOLOv8 · ByteTrack · Roboflow · CustomTkinter · Multithreading

- Solved identity fragmentation: a custom ObjectTracker maps transient ByteTrack IDs to session-permanent UIDs that survive occlusions and frame exits.
- Fine-tuned YOLOv8 on a custom Roboflow-labelled dataset; managed the full lifecycle from data preparation to deployment.
- Producer–consumer architecture decouples inference from a branded desktop GUI (stable 30 Hz) with alerts, live search and Excel export.

Industrial Computer Vision Suite (Dual-Module)

Python · OpenCV · CustomTkinter · Threading

- Unified an HSV colour (7 profiles) & shape classifier (Douglas–Peucker, circularity and aspect-ratio heuristics) and a real-time QR/barcode decoder with perspective rectification under one GUI launcher.
- Offloaded vision loops to daemon threads with a bounded thread-safe queue, keeping the dashboard at 30+ FPS; telemetry exported to Excel-ready CSV.

Retail Bulk-Produce Classifier

TensorFlow / Keras · CNN · Streamlit

- Designed a custom 5-stage hierarchical CNN on Fruits-360, reaching **97% macro F1** across 6 core produce categories; deployed as an interactive Streamlit dashboard.

Autonomous Mobile Robots — Line Follower & Maze Solver

Arduino · PID Control · CoppeliaSim · SolidWorks · GitHub

- Designed, built and tuned a high-speed PID line-follower with multi-sensor detection; competed in the course robotics competition.
- Simulated line-following and maze-solving robots in CoppeliaSim (sensor integration, obstacle avoidance, path planning) before hardware deployment.

Wireless Gesture-Controlled 4WD Robot

Arduino Uno/Nano · MPU6050 · HC-05 Bluetooth · L298N

- Translated IMU hand tilts into real-time motion commands over a master–slave Bluetooth link, with calibrated thresholds and Li-ion 18650 power management.

LEADERSHIP & VOLUNTEERING

Branch Treasurer (Executive Board) | IEEE Benha University Student Branch — Sep 2025 – Aug 2026

- Managed branch finances, budgeting and funding allocations for all activities, including mega events TURBO CUP and LOST.
- Led three days of recruitment interviews, screening and evaluating 750+ applicants across committees.
- Supervised the first-ever launch of the branch's IEEE Computer Society Chapter and SIGHT Group.
- Managed Let's Embark (4th ed., record attendance); secured sponsorships for LOST; supervised ResearchX with RAS; launched the first Tachtech Ramadan initiative.

Fundraising Committee Member | IEEE Benha University Student Branch — Oct 2023 – Sep 2024

- Secured and managed sponsorships for SPIRIT 2024; Operations Manager of Mega Event 2024 (RoboSoccer, with the RAS Chapter).
- **Awarded Best Member of the Fundraising Committee (2024).**

Head of Business Development Committee | Banha Rally — Apr 2025 – Sep 2025

IEEE Ambassadorships

IEEE Region 8 Entrepreneurship Committee
Sep 2026 – Present

IEEE Sensors Council Egypt Chapter
Jul 2026 – Present

IEEEXtreme 20.0 — SB Ambassador
Jul – Nov 2026

TSYP 14.0 Congress, Tunisia
Jul 2026 – Present

IEEE CS R8 Chapter Champions League
Oct 2026 – Present

TechX 2K26 — Campus Ambassador
Oct – Nov 2026

COMPETITIONS, HACKATHONS & PROGRAMS

- **IEEE AESS Sustainability Hackathon (AESH) 2026** — inter-chapter regional hackathon (Tunisia, Egypt, Jordan); recognized for technical innovation.
- **IEEE SIGHT Egypt — HIIVE Program** — Certificate of Achievement for human-centric solutions to local community challenges.
- **NASA Space Apps Challenge, Cairo 2024** — co-built a data-driven exoplanet explorer website, YouTube channel, Kahoot! quiz and board game.
- **IEEE Entrepreneurship Week Egypt 2026** (Elsewedy University of Technology) — ideation, market validation and pitching.
- **TechX at E-JUST** — represented the IEEE Benha Student Branch.
- **GHOSTECH 2022** — 5-day hardware & software workshops and competition.

COURSES & CERTIFICATIONS

- **Artificial Intelligence Master Class (100 hrs)** — Creativa Hub Benha (MCIT · ITIDA · TIEC), Jul–Aug 2026 · **Capstone score: 100%**
- **Artificial Intelligence** — National Telecommunication Institute (NTI), 2025 · Orange data analysis; portfolio-website capstone
- **AI & Data Analysis Workshop** — IEEE Suez Student Branch
- **Mechanical Design with SolidWorks** — Udemy, 2024
- **Industrial Automation** — LinkedIn Learning, 2024
- **Building Business Reports Using Power BI** — 365 Data Science, 2023